# The Timing of Early-Life Shocks and Adolescent Mental Health

October 2026

# Do the effects of an earthquake experienced at ages 0–4 persist?




**Shock** 27 February 2010 (27-F), magnitude 8.8

**Exposed** Children aged 0–4

**Effects of early-childhood shocks**

Birth weight (Torche, 2011)

Cognitive and non-cognitive development (Gillmore, 2026)

Height and learning (Andrabi, Daniels, and Das, 2023)

Adult mental health (Persson and Rossin-Slater, 2018)

**This paper**

Mental health in adolescence, ages 14–18

# Does it matter at what age children live through the shock?

> "The effects of stress on the brain depend critically on the timing (age of onset and duration). When stress occurs early in life it can have profound and lasting effects on brain organization and function." (Gee and Casey, 2015, p. 184)

# Potential mechanisms

| | Channels |
|---|---|
| **Socioeconomic and family environment** | The earthquake reduces income, damages housing, and increases conflict at home (Conger and Donnellan, 2007; Almond, Currie, and Duque, 2018) |
| **Transmission of maternal stress** | Infants regulate their stress through the mother. If she is stressed, she passes it on through daily contact (Persson and Rossin-Slater, 2018) |

# Data



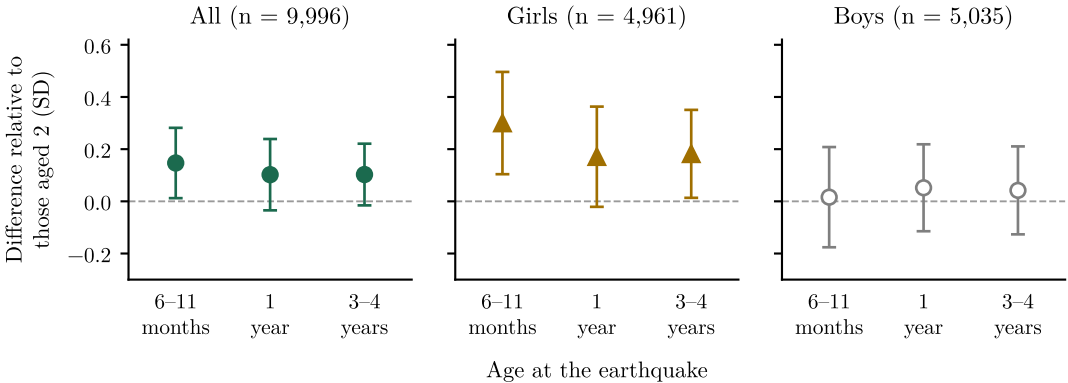

| | |
|---|---|
| Survey | ELPI, Early Childhood Longitudinal Survey, Chile |
| Sample | 9,996 adolescents born in 2006–2009; 4,961 girls |
| Respondents | The caregiver in every round; the child in 2017 and 2024 |

# Econometric specification

$$Y_{imc} = \sum_{a \neq 2} \beta_a \left( \text{Age}_{ai} \times \text{Affected}_m \right) + \sum_{a \neq 2} \gamma_a \, \text{Age}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}$$

| | |
|---|---|
| $Y_{imc}$ | Mental health at ages 14–18, PHQ-4 |
| $\text{Age}_{ai}$ | Age at the earthquake: 6–11 months, 1 year, or 3–4 years; reference, 2 years |
| $\text{Affected}_m$ | Municipality in the zone of destructive damage |
| $\psi_m$, $\theta_c$ | Municipality and birth-cohort fixed effects |
| $X_i$ | Child and household controls |

# Effect on mental health

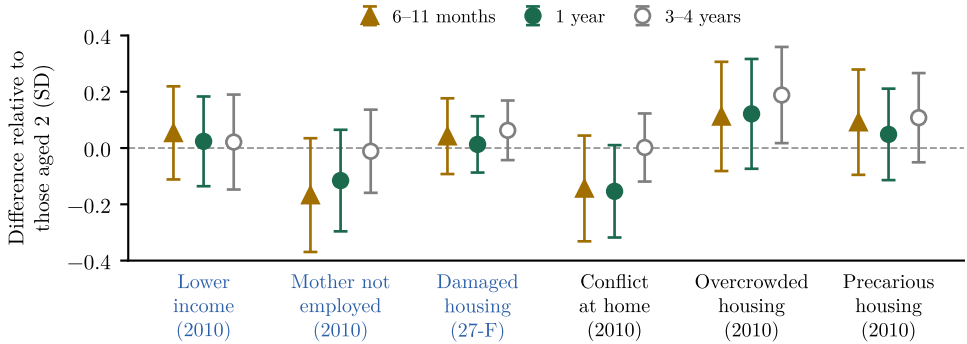



Mental health is measured with the PHQ-4 (Kroenke et al., 2009), four questions answered by the adolescent on symptoms of anxiety and depression (nervousness, worry, loss of interest, and feeling down). Each point is the difference between affected and unaffected municipalities, with 95% CI.

▸ Table A1

# Socioeconomic and family environment does not explain the gap

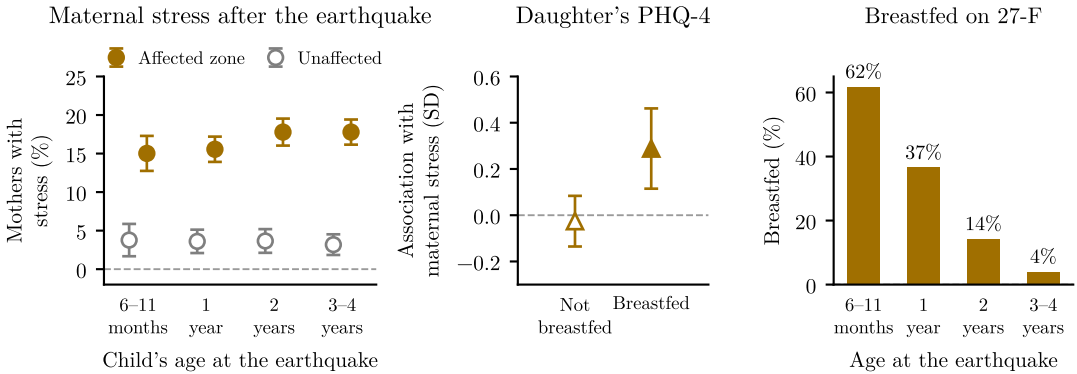
The earthquake worsened the environment (Gillmore, 2026), but with no differences by age




Measured in 2010, one to nine months after the earthquake. In blue, the mechanisms in Gillmore (2026). Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack (*mediagua*) or a lent or shared dwelling. Girls, relative to those aged 2; 95% CI.

▸ Table A2

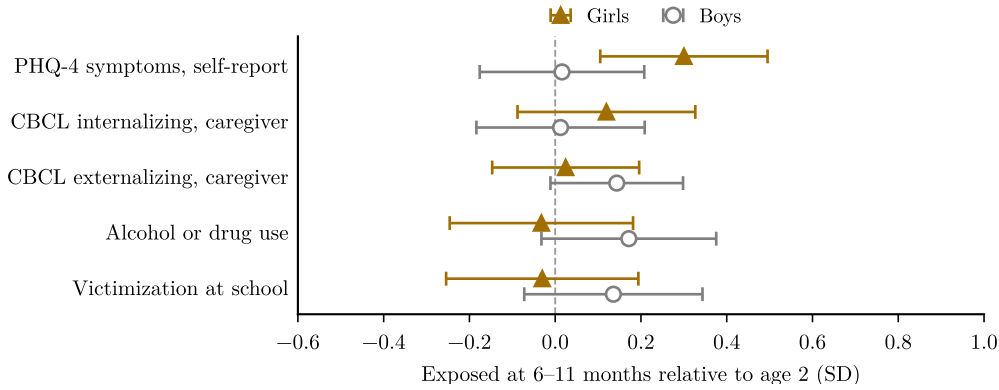
# Maternal stress and daughters' mental health by breastfeeding




Active breastfeeding on 27-F serves as a proxy for regulatory dependence and biochemical and behavioral proximity to the mother. Girls, with fixed effects and controls, also for the mother's education and for age; 95% CI.

▸ Table A3 ▸ Table A4

# Among boys, only weak signs in behavior and substance use




▸ Table A5 ▸ Table A6

# Findings and open questions

## What the data suggest

- Girls exposed at 6–11 months have worse mental health than those exposed at age 2, by 0.30 SD on the PHQ-4.
- The socioeconomic and family environment changed similarly at all ages and does not explain this difference.
- Maternal stress is associated with worse mental health only among daughters who were breastfed, 62% of those exposed at 6–11 months.

## What remains open

- What explains the effect at ages 3–4, if it operates neither through the socioeconomic and family environment nor through maternal stress?

  **Hypothesis:** at that age children can already remember consciously what they experience, and the memory of the earthquake could affect their mental health.

# References


Almond, Douglas, Janet Currie, and Valentina Duque. 2018. “Childhood Circumstances and Adult Outcomes: Act II.” ***Journal of Economic Literature*** 56 (4): 1360–1446.

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Kroenke, Kurt, Robert L. Spitzer, Janet B. W. Williams, and Bernd Löwe. 2009. “An Ultra-Brief Screening Scale for Anxiety and Depression: The PHQ-4.” ***Psychosomatics*** 50 (6): 613–621.

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## Table A1—Effect on Mental Health

| | PHQ-4 (SD) | | |
|---|---|---|---|
| | (1) All | (2) Girls | (3) Boys |
| 6–11 months × Affected | 0.147 | 0.300 | 0.016 |
| | (0.069) | (0.100) | (0.098) |
| 1 year × Affected | 0.102 | 0.171 | 0.052 |
| | (0.070) | (0.098) | (0.085) |
| 3–4 years × Affected | 0.103 | 0.182 | 0.042 |
| | (0.060) | (0.086) | (0.086) |
| Mean of outcome, aged 2 | 0.017 | 0.233 | −0.196 |
| Observations | 9,996 | 4,961 | 5,035 |
| Municipality, cohort, and age-group FE | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes |

*Notes:* The table reports the difference between affected and unaffected municipalities in the PHQ-4 score (standardized) for each age at the earthquake, relative to those aged 2. All regressions include municipality, birth-cohort, and age-group fixed effects and controls for age in months, sex, the mother's age and education, household size, and rural residence (2010), the mother's number of children, and family history of mental illness (2012), and use ELPI sampling weights. Mean of outcome is the weighted mean for those aged 2 at the earthquake. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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# Table A2—The Socioeconomic and Family Environment

| | Mechanisms in Gillmore (2026), SD | | | Other household conditions, SD | | |
|---|---|---|---|---|---|---|
| | (1) Lower income | (2) Mother not employed | (3) Damaged housing | (4) Conflict at home | (5) Overcrowded housing | (6) Precarious housing |
| 6–11 months × Affected | 0.054 | −0.167 | 0.042 | −0.144 | 0.112 | 0.092 |
| | (0.084) | (0.103) | (0.069) | (0.096) | (0.099) | (0.095) |
| 1 year × Affected | 0.024 | −0.116 | 0.013 | −0.154 | 0.121 | 0.049 |
| | (0.081) | (0.092) | (0.051) | (0.084) | (0.100) | (0.083) |
| 3–4 years × Affected | 0.021 | −0.011 | 0.063 | 0.002 | 0.188 | 0.108 |
| | (0.086) | (0.075) | (0.054) | (0.062) | (0.087) | (0.081) |
| Mean of outcome, aged 2 | 0.028 | −0.008 | −0.037 | −0.045 | −0.008 | −0.013 |
| Observations | 4,479 | 4,926 | 4,424 | 4,702 | 4,962 | 4,961 |
| Municipality, cohort, and age-group FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |

*Notes:* Girls. Each column is a standardized index of the household environment; higher is worse. Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack or a lent or shared dwelling. Specification as in Table A1. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

## Table A3—Maternal Stress, Breastfeeding, and Daughters' Mental Health

| | Daughters' PHQ-4 (SD) | | |
|---|---|---|---|
| | (1) Baseline | (2) Mother's SES | (3) Mother's SES and infancy |
| Maternal stress × not breastfed | −0.026 | −0.005 | −0.004 |
| | (0.056) | (0.063) | (0.064) |
| Maternal stress × breastfed | 0.288 | 0.280 | 0.286 |
| | (0.089) | (0.096) | (0.112) |
| Difference, breastfed − not breastfed | 0.314 | 0.285 | 0.290 |
| | (0.108) | (0.102) | (0.110) |
| Mean of outcome, mother without stress | 0.206 | 0.204 | 0.204 |
| Observations | 4,409 | 4,377 | 4,377 |
| Municipality, cohort, and age group × breastfed FE | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes |
| Stress and breastfed × mother's education and employment | No | Yes | Yes |
| Stress × 6–11 months | No | No | Yes |

*Notes:* Girls. Maternal stress equals one if the mother reported stress, distress, or anxiety after the earthquake (2012); breastfed equals one if the child was still breastfed on 27-F (2012 report). Column 1 is the estimate in the figure; column 2 lets stress and breastfeeding vary with the mother's education and employment (2010); column 3 also lets stress vary at 6–11 months. Controls and weights as in Table A1. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

## Table A4—Maternal Stress and Breastfeeding by Age at the Earthquake

| | Mothers with stress (%) | | Breastfed (%) |
|---|---|---|---|
| | (1) Affected | (2) Unaffected | (3) On 27-F |
| 6–11 months | 15.0 | 3.8 | 61.8 |
| | (1.2) | (1.1) | |
| 1 year | 15.6 | 3.6 | 36.6 |
| | (0.8) | (0.8) | |
| 2 years | 17.8 | 3.7 | 14.3 |
| | (0.9) | (0.8) | |
| 3–4 years | 17.8 | 3.2 | 4.1 |
| | (0.8) | (0.7) | |
| Children | 6,776 | 2,134 | 8,880 |

*Notes:* Unweighted means, with standard errors in parentheses, by the child's age at the earthquake. Columns 1 and 2 report the share of mothers who reported stress, distress, or anxiety after the earthquake (2012), in affected and unaffected municipalities; column 3, the share of children still breastfed on 27-F.

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# Table A5—Other Outcomes, Girls

| | Adolescent report (SD) | | | Caregiver report, CBCL (SD) | |
|---|---|---|---|---|---|
| | (1) PHQ-4 | (2) Substance use | (3) Victimization | (4) Internalizing | (5) Externalizing |
| 6–11 months × Affected | 0.300 | −0.032 | −0.030 | 0.119 | 0.024 |
| | (0.100) | (0.109) | (0.114) | (0.106) | (0.087) |
| 1 year × Affected | 0.171 | 0.026 | 0.044 | 0.154 | 0.100 |
| | (0.098) | (0.096) | (0.081) | (0.120) | (0.107) |
| 3–4 years × Affected | 0.182 | −0.094 | 0.081 | 0.042 | −0.082 |
| | (0.086) | (0.083) | (0.092) | (0.095) | (0.083) |
| Mean of outcome, aged 2 | 0.233 | 0.030 | −0.073 | 0.104 | 0.056 |
| Observations | 4,961 | 4,961 | 4,539 | 4,748 | 4,748 |
| Municipality, cohort, and age-group FE | Yes | Yes | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes | Yes | Yes |

*Notes:* Girls. Each column is a separate regression with the specification of Table A1. The adolescent reports the PHQ-4, substance use (any alcohol, tobacco, cannabis, or other drug in the last 12 months), and peer victimization and class climate; the caregiver reports internalizing and externalizing problems (CBCL). All measured in 2024 and standardized; higher is worse. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

## Table A6—Other Outcomes, Boys

| | Adolescent report (SD) | | | Caregiver report, CBCL (SD) | |
|---|---|---|---|---|---|
| | (1) PHQ-4 | (2) Substance use | (3) Victimization | (4) Internalizing | (5) Externalizing |
| 6–11 months × Affected | 0.016 | 0.172 | 0.136 | 0.012 | 0.144 |
| | (0.098) | (0.104) | (0.106) | (0.100) | (0.079) |
| 1 year × Affected | 0.052 | −0.064 | 0.112 | −0.063 | 0.053 |
| | (0.085) | (0.089) | (0.086) | (0.099) | (0.092) |
| 3–4 years × Affected | 0.042 | −0.031 | 0.098 | 0.010 | 0.126 |
| | (0.086) | (0.079) | (0.094) | (0.099) | (0.090) |
| Mean of outcome, aged 2 | −0.196 | −0.027 | −0.063 | −0.084 | −0.033 |
| Observations | 5,035 | 5,034 | 4,637 | 4,804 | 4,804 |
| Municipality, cohort, and age-group FE | Yes | Yes | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes | Yes | Yes |

*Notes:* Boys. Outcomes and specification as in Table A5. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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